The G+EM Orthogonal Triad: Gravity as the Longitudinal Component of Electromagnetic Radiation

John Lehew

Independent Researcher
Email: john.lehew@omega137.com | ORCID: 0009-0007-5895-7787

February 22, 2026

Keywords: Gravitoelectromagnetism, General Relativity, Quantum Electrodynamics, Stress-Energy Tensor, Topological Compactification, Pair Production.

Abstract

If matter is generated from light via pair production, the gravitational signature of mass cannot spontaneously emerge upon particle creation; it must be an intrinsic, constitutive component of electromagnetic radiation. This paper establishes the Gravity, Electricity, and Magnetism (GEM) triad, demonstrating that the transverse electric and magnetic forces on charge are inextricably coupled to a third, longitudinal force: the gravitational metric tension on mass-energy. By identifying the electromagnetic stress-energy tensor ($T^{\text{EM}}_{\mu\nu}$) as the rank-2 bridge to spacetime curvature, we establish that the vacuum constants ε_0, μ_0, and G are not independent phenomenological values, but the permittivity, permeability, and metric stiffness of a single continuous medium.

We model the conversion of radiation to matter as a topological compactification: a unitary rotation of the wavevector ($k \to i\kappa$) that compactifies longitudinal propagation into static inertia. Crucially, this gravitational component is functionally necessary for photon-matter interaction. The quantum requirement for directed momentum ($\mathbf{k} \neq 0$) is identically the condition that sources spacetime curvature ($T^{\text{EM}}_{0i} \neq 0$). A photon stripped of its capacity to source gravity would carry no momentum and could not be absorbed.

1 Introduction

Light bends around the Sun.

This fact, confirmed by Eddington in 1919 [4], rests on a physical reality that the Einstein–Maxwell equations [2, 3] have encoded for over a century: light couples to spacetime curvature through the stress-energy tensor. Standard physics treats this coupling as a macroscopic metric response, one source term among many feeding into $T_{\mu\nu}$. In the classical limit, the gravitational force between two objects predicts zero force for a massless photon:

$$F = G\frac{m_1 m_2}{r^2}\,. \qquad (1)$$

Light should travel in perfectly straight lines, indifferent to the cosmic mass distribution.

It does not. General relativity resolves this by establishing that light follows geodesics in curved spacetime, because spacetime curvature is sourced by the stress-energy tensor ($T_{\mu\nu}$), specifically energy density and momentum flux, of which rest mass is merely one highly localized expression.

This mathematical resolution is universally accepted, yet its deeper physical implication remains underappreciated. Standard physics recognizes that light *sources* gravity through the Einstein–Maxwell equations. What it does not do is recognize the gravitational contribution as an **intrinsic, foundational property of the quantum wave itself**, equivalent in status to the electric and magnetic fields. The electric field oscillates. The magnetic field oscillates. The energy-momentum tensor contributes to spacetime curvature, and that longitudinal metric tension acts as gravitational charge, creating attraction toward the beam, binding light to the fabric of spacetime.

Standard physics sees two intrinsic components in light: E and M. We see three: E, M, and G.

These form an orthogonal triad we call **GEM**. Maxwell unified E and M in 1865 by demonstrating $c^2 = 1/(\varepsilon_0\mu_0)$. We argue that G completes the unification, not as a speculative addition to the Standard Model, but as a structural reframing of established physics.

The key insight concerns the vacuum itself. The electromagnetic properties of empty space are characterized by ε_0 (permittivity) and μ_0 (permeability). However, the vacuum possesses a third measurable property: its resistance to geometric curvature. In the Einstein–Hilbert action [3], the coefficient $c^4/(16\pi G)$ dictates how strongly spacetime resists deformation by energy. It is the **metric stiffness** of the vacuum (distinct from the intrinsic vacuum energy density described by the cosmological constant Λ). We propose that ε_0, μ_0, and G are not independent constants describing separate physics. They are three fundamental properties of a single continuous medium. Light is the primary signal of this medium; gravity is not a separate force impressed upon it, but the consequence of the medium's own stiffness reacting to electromagnetic flux. The exact algebraic geometry linking these vacuum constants, including the closed-form derivation of G and the fine-structure constant (α), is detailed in a companion framework [12].

A natural objection arises: if the GEM triad is already encoded in the stress-energy tensor of general relativity, what does this paper contribute? The answer is twofold. First, historical physics treats the electromagnetic contribution to $T_{\mu\nu}$ as simply one source among many (matter, fluid, dark energy) without recognizing its primacy. Electromagnetism is the only massless, infinite-range force that natively generates gravity. Furthermore, through pair production, matter itself is generated from confined electromagnetic radiation, meaning all gravitational mass ultimately traces its origin back to GEM radiation. Second, this structural reframing generates concrete, falsifiable predictions: if ε_0, μ_0, and G are properties of a single unified medium, then gravitational and electromagnetic waves must propagate at identical velocities, and light's gravitational content must exhibit tensor structure (not scalar). Both predictions are confirmed by existing experiments (Sections III–V).

The reframing is subtle. Its implications are absolute.

2 The Geometric Evidence: r^2 in the Weak-Field Limit

2.1 Three Forces, One Shared Space

In the classical, weak-field macroscopic limit, the electric, magnetic, and gravitational forces all share an identical inverse-square mathematical structure ($F \propto 1/r^2$). This dependence is an absolute consequence of geometry: any conserved influence radiating outward in three-dimensional space dilutes across spherical surfaces of area $4\pi r^2$. While this shared inverse-square geometry is a necessary consequence of three-dimensional flux conservation, it fundamentally establishes that all three fields are expressions projected through the exact same spatial metric. The crucial question: are K_e, K_m, and G independent phenomenological constants, or are they algebraically linked constraints of the same underlying metric medium?

2.2 Maxwell's Precedent

In 1865, Maxwell unified electricity and magnetism [1] by deriving a wave equation with a phase velocity of:

$$c = \sqrt{\frac{K_e}{K_m}} = \frac{1}{\sqrt{\varepsilon_0 \mu_0}}\,. \tag{2}$$

Using laboratory measurements (capacitors for ε_0, inductors for μ_0), this calculated speed perfectly matched the astronomically measured speed of light. This was a prediction, not a definition. The profound implication was that electricity and magnetism are not separate forces, but orthogonal expressions of a single field, with their coupling constants locked together by the geometry of the vacuum:

$$c^2 = \frac{K_e}{K_m}\,. \tag{3}$$

2.3 The Question for Gravity

If K_e and K_m are inherently algebraically related through c^2, what about G? Standard physics treats G as an orphaned, empirical parameter. We will demonstrate that three independent lines of experimental evidence suggest otherwise, proving that G is functionally inextricable from the electromagnetic properties of light.

3 The Kinematic Identity Constraint: Same Speed

3.1 LIGO/Fermi 2017

On August 17, 2017, the LIGO-Virgo collaboration [7] detected gravitational waves from a neutron star merger 130 million light-years distant. The Fermi telescope [8] detected a gamma-ray burst from the same event 1.7 seconds later.

Both signals traveled 1.2×10^{24} meters across the universe. Their arrival times constrain the fractional difference in fundamental vacuum phase velocities to an extraordinarily tight asymmetric bound:

$$-3 \times 10^{-15} \leq \frac{v_{\text{gravity}} - c_{\text{light}}}{c_{\text{light}}} \leq +7 \times 10^{-16}\,. \tag{4}$$

3.2 The Significance of Kinematic Identity

If gravity were an inherently independent phenomenon from electromagnetism, this equality would be a staggering coincidence demanding explanation. Why should a tensor force sourced by mass-energy propagate at a velocity strictly defined by the electromagnetic vacuum constants (ε_0 and μ_0)?

The constraint is absolute. Both signals traversed 130 million light-years of spacetime. While the gamma rays experienced a minute 1.7-second delay due to astrophysical breakout times and slight refractive interactions with interstellar plasma (which gravitational waves pass through undisturbed), their vacuum phase velocities are virtually identical. If the physical metric mechanism propagating gravity were distinct from the mechanism propagating vector electromagnetism, there is no physical reason their fundamental vacuum speeds would perfectly match to fifteen decimal places. Even a fractional divergence of one part in a billion would have resulted in months of temporal separation over 1.2×10^{24} meters.

Under the GEM framework, no coincidental explanation is needed. Both propagate at c because both are components of the same unified phenomenon. The speed c is not merely the "speed of light"; it is the geometric phase velocity of all GEM interactions. Gravitational waves propagate at precisely $c = 1/\sqrt{\varepsilon_0\mu_0}$ because their propagation is governed by the exact same vacuum constraints that govern light. The speed is not merely shared. It is inherited from the same unified metric.

The equality is not coincidence. It is identity.

4 The Energetic Evidence: Light Gravitates

4.1 The Paradox Sharpened

Return to Equation (1). With $m_1 = 0$, the classical predicted force is zero.

Yet observation confirms:

- Light bends around massive objects (Eddington, 1919 [4])
- Light cannot escape black holes
- Light loses energy climbing gravitational wells (Pound–Rebka, 1960 [5])

Three phenomena. All measured. All impossible if light exerts and experiences zero gravitational interaction. The resolution is not that $F = 0$ is wrong. The resolution is that the effective gravitational mass is non-zero when energy is present.

4.2 Energy Is Gravitational Charge

For a photon, the stress-energy tensor $T_{\mu\nu}$ is constructed *entirely* from electromagnetic quantities: field energy, momentum flux, and radiation pressure. There is no separate "gravitational substance" inside the photon. The gravitational field of light is generated by its electromagnetic content alone. The G in light is not optional. It is mandatory.

Higher frequency translates directly to greater gravitational mass equivalence per photon ($m_{\text{grav}} = h\nu/c^2$), and a brighter beam possesses greater energy density ($u = \varepsilon_0 E_0^2$), generating a proportionally larger gravitational metric response.

4.3 Experimental Confirmation

Eddington (1919) [4], Bidirectional Coupling: Starlight grazing the Sun deflects by 1.75 arcseconds, exactly twice the Newtonian prediction for an object moving at c with mass $m = E/c^2$. The doubling confirms that light couples to the full geometric curvature of spacetime, not just a Newtonian scalar potential.

But the first-principles implication runs deeper. The conservation of momentum demands that interactions are never unilateral. If the Sun's gravitational field alters the momentum vector of a photon, the photon must exert a reciprocal gravitational influence on the Sun. The photon is not a passive object being deflected; it is an active gravitational source, pulling back on the star that bends its path. The Eddington result does not merely show that light *responds* to gravity. It proves, by conservation of momentum, that light *generates* gravity.

Pound–Rebka (1960) [5], Direct Energy Exchange: Gamma rays climbing 22.5 meters in Earth's gravitational field shift in frequency:

$$\frac{\Delta\nu}{\nu} = \frac{gh}{c^2}\,. \tag{5}$$

This exact transaction is a direct manifestation of the Einstein Equivalence Principle applied to massless radiation. Photons lose energy climbing out of gravity wells and gain energy falling in. This is not an indirect geometric effect. It is a direct, measurable exchange of currency between the electromagnetic and gravitational degrees of freedom. The photon pays a gravitational toll out of its own internal electromagnetic clock ($E = h\nu$). The frequency decreases; the gravitational potential energy increases. The transaction is exact, governed by c^2.

If electromagnetism and gravity were fundamentally independent systems, separate forces that merely coexist in the same spacetime, they could not seamlessly exchange energy quanta with such algebraic precision. The Pound–Rebka experiment demonstrates that the electromagnetic oscillation of a photon is inherently coupled to the gravitational potential of the metric it occupies. They are not independent accounts. They draw from the exact same physical balance.

5 Why Light Doesn't Attract Light: Null Causal Isolation

If photons carry a gravitational tensor component, why doesn't a beam of light gravitationally attract itself? Over 100 light-years of travel, parallel photons should converge, smearing starlight into clumps. They do not.

5.1 The Tolman–Ehrenfest–Podolsky Result and Null Causal Isolation

This apparent contradiction was resolved mathematically by Tolman, Ehrenfest, and Podolsky in 1931 [6], but its geometric meaning is often misinterpreted as a mere "cancellation" of forces. In the GEM framework, the lack of attraction between parallel photons is not a cancellation; it is an absolute consequence of **Null Causal Isolation**.

For two photons traveling exactly parallel in the $+z$ direction, their relative invariant mass is strictly zero: $s = (p_1 + p_2)^2 = 0$. Because both propagate on parallel null geodesics at velocity c, no signal, gravitational or otherwise, can traverse the transverse gap between them without exceeding the speed of light.

Furthermore, the gravitational component of the GEM wave is a longitudinal metric tension aligned exclusively with the Poynting vector ($\mathbf{S}$). For parallel photons, these longitudinal G-components are geometrically "out of phase" with their relative transverse spatial separation. They do not balance each other out; they are fundamentally **blind** to one another. The metric tension exists, but they cannot "see" it because no causal bridge can cross their parallel trajectories.

If the photons' trajectories diverge by even an infinitesimal angle θ, their invariant mass becomes non-zero ($s > 0$), their causal horizons overlap, and the longitudinal metric tension ceases to be perfectly out-of-phase, resulting in a calculable, non-zero gravitational interaction. The strict null isolation applies only to mathematically perfect parallel alignment.

The fact that light does not attract parallel light is not evidence against the G-component; it is the ultimate proof that gravity in light obeys the strict geometric limits of tensor propagation.

6 The Experimental Chain of Custody

The preceding four experimental results, taken together, form a closed logical chain. Each result, individually, is explained by standard physics. Their *combination* is what reveals the inescapable GEM structure.

Experiment	**What It Proves**	**Escape Route Closed**
Eddington (1919) [4]	Light generates gravity (bidirectional coupling)	"Light is just a passive test particle"
Pound–Rebka (1960) [5]	EM and gravity exchange energy directly	"They are independent systems that don't interact"
LIGO/Fermi (2017) [7, 8]	Light and gravity share the vacuum propagation medium	"Same speed is coincidence from Lorentz invariance"
TEP (1931) [6]	Light's gravity is a structured tensor, not scalar energy	"Gravity in light is just generic $E = mc^2$"

Table 1: The experimental chain of custody for the GEM triad.

The experimental chain of custody is complete. Light does not merely *experience* gravity. Light *generates* gravity, *exchanges energy* with gravity, *propagates at gravity's fundamental speed*, and carries gravity as a *structured tensor component* of its stress-energy. The question is no longer whether gravity is part of light. The question is why we ever considered them separate.

7 The Rank-2 Bridge: From Vector Electromagnetism to Tensor Gravity

7.1 The Spin Mismatch and the Quantum Resolution

The primary technical objection to the GEM unification is the **spin mismatch**. Electromagnetism is a vector phenomenon, mediated in the quantum regime by a spin-1 boson (the photon). Gravity is a tensor phenomenon, represented by the rank-2 metric tensor $g_{\mu\nu}$ (and mediated by a spin-2 state). How can a spin-1 vector phenomenon fundamentally generate a spin-2 tensor geometry?

The resolution exists at the intersection of Quantum Field Theory and General Relativity. Gravity does not couple to the vector potential A_μ directly; it couples to the electromagnetic stress-energy tensor, $T_{\mu\nu}^{\mathrm{EM}}$. In QFT, this tensor is a composite operator constructed from bilinear products of the spin-1 field. When we take the tensor product of two spin-1 fields, the Clebsch–Gordan decomposition of the angular momentum states yields [11]:

$$1 \otimes 1 = 2 \oplus 1 \oplus 0\,. \tag{6}$$

This demonstrates that the bilinear product of a spin-1 field natively decomposes into a scalar (spin-0, the trace), an antisymmetric vector (spin-1), and a symmetric, traceless rank-2 tensor (spin-2). The electromagnetic stress-energy tensor $T_{\mu\nu}^{\mathrm{EM}}$ is precisely this symmetric spin-2 representation. The quantum mechanical capacity to source a spin-2 metric response is not added to the photon; it is an intrinsic geometric consequence of the field's own bilinear self-interaction.

7.2 The Classical Manifestation

Macroscopically, via the correspondence principle and normal-ordering of the quantum fields, this quantum spin-2 operator manifests as the classical electromagnetic stress-energy tensor. Constructed from the antisymmetric Faraday field tensor $F_{\mu\nu}$, it takes the form:

$$T_{\mu\nu}^{\mathrm{EM}} = \frac{1}{\mu_0}\left(F_{\mu\alpha}F_\nu{}^\alpha - \frac{1}{4}g_{\mu\nu}F_{\alpha\beta}F^{\alpha\beta}\right)\,. \tag{7}$$

This rank-2 symmetric tensor matches the exact rank and symmetry of the Einstein tensor $G_{\mu\nu}$. It couples directly to gravity through the field equations ($G_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}^{\mathrm{EM}}$). The absolute bridge between vector electromagnetism and tensor gravity is $T_{\mu\nu}^{\mathrm{EM}}$ itself.

7.3 The Poynting Vector: Momentum, Gravity, and Absorption

The physical content of this bridge is most transparent through the Poynting vector [10], $\mathbf{S} = (\mathbf{E}\times\mathbf{B})/\mu_0$. Its components are identically the time-space components of the stress-energy tensor:

$$T_{0i}^{\mathrm{EM}} = \frac{S_i}{c}\,. \tag{8}$$

This is an algebraic identity, not an approximation. The Poynting vector is also the *sole carrier* of electromagnetic momentum density:

$$\mathbf{g} = \frac{\mathbf{S}}{c^2} = \varepsilon_0(\mathbf{E}\times\mathbf{B})\,. \tag{9}$$

If $\mathbf{S} = 0$, the field carries zero momentum, and neither absorption nor emission can conserve momentum. The proof that every absorbable photon must gravitate proceeds in four steps:

1. **Conservation of momentum**: Absorption and emission require nonzero directed photon momentum.
2. **Maxwell's equations**: Electromagnetic momentum is carried exclusively by $\mathbf{S}$.
3. **Stress-energy identity**: $S_i/c = T_{0i}^{\mathrm{EM}}$ (Equation 8).
4. **Einstein field equations**: Nonzero $T_{\mu\nu}^{\mathrm{EM}}$ necessarily sources spacetime curvature.

$$\text{Absorption/emission} \implies \mathbf{S} \neq 0 \implies T_{0i}^{\mathrm{EM}} \neq 0 \implies \text{gravitational sourcing}$$

The "G" in a GEM wave is not a new field patched onto electromagnetism. It is the mandatory metric response to the longitudinal spin-2 energy flux already present in every electromagnetic wave. Wherever $\mathbf{E}$ and $\mathbf{B}$ oscillate, $\mathbf{S}$ flows, $T_{\mu\nu}^{\mathrm{EM}}$ is nonzero, and the metric must curve. Every photon capable of being absorbed or emitted necessarily carries a gravitational component.

7.4 The GEM Action Principle

The total action of a spacetime containing electromagnetic fields is:

$$\mathcal{S}_{\mathrm{Total}} = \int \left(\frac{c^4}{16\pi G} R - \frac{1}{4\mu_0} F_{\mu\nu} F^{\mu\nu} \right) \sqrt{-g}\, d^4x \,. \tag{10}$$

The GEM framework identifies these as two material properties of a **single monolithic medium**. The coefficient $c^4/(16\pi G)$ is the **metric stiffness** of the vacuum; $1/(4\mu_0)$ governs field propagation through that exact same vacuum. The feedback loop, where the EM field curves spacetime and spacetime curvature guides the EM field, is the internal geometric tension of the GEM triad.

If GEM is correct, these coefficients are not independent parameters but coupled properties of a single geometric fabric.

8 The Gravitational Content of Light

8.1 Direction as Gravitational Content

The Poynting vector encodes the photon's direction of travel. This information is intrinsic and local to the photon; no external field or cosmic ledger tracks which direction each photon propagates. The photon's "I am moving in direction $\hat{k}$" is physically encoded in $\mathbf{S}$, which is identically T_{0i}^{EM}, which is the gravitational source term.

While a classical electromagnetic field with a net $\mathbf{S} = 0$ can exist as a standing wave (a macroscopic superposition), at the quantum level, a single freely-propagating photon cannot exist in a state of $\mathbf{S} = 0$. Because its energy is strictly defined by its momentum ($E = \hbar c|\mathbf{k}|$), removing its directed momentum removes its energy entirely. It does not vibrate in place; it ceases to exist. The G component of light is not merely what makes it gravitate. It is what makes it *go*.

This resolves a question that the standard framework leaves implicit: why does every physical photon carry gravitational content? Because every physical photon has nonzero frequency ($\omega \neq 0$), therefore nonzero wavevector ($\mathbf{k} \neq 0$), therefore nonzero momentum ($\mathbf{p} = \hbar\mathbf{k} \neq 0$), therefore nonzero Poynting vector ($\mathbf{S} \neq 0$), therefore nonzero T_{0i}^{EM}, therefore nonzero gravitational sourcing. A zero-momentum photon would have zero energy, zero frequency, and would not exist. The gravitational component is not an incidental property of light. It is a necessary condition for light to be light.

8.2 Phase Synchronization and Absorption

The wavevector $\mathbf{k}$ carries more than direction; it encodes the photon's spatial phase via $e^{i\mathbf{k}\cdot\mathbf{r}}$. For a photon to be absorbed by an atom, this phase must satisfy the resonance condition with the target electron's wavefunction. The photon must arrive not only at the correct location but with the correct phase to constructively interfere with the electron's transition dipole moment. If the phase relationship is destructive, absorption does not occur; the photon passes through.

This phase coherence is inseparable from the gravitational content. In standard QED, absorption is governed by the interaction Hamiltonian ($H_{\mathrm{int}} \propto \mathbf{A} \cdot \mathbf{p}$), where the spatial phase $e^{i\mathbf{k}\cdot\mathbf{r}}$ is explicitly housed within the mode expansion of the vector potential $\mathbf{A}$. This strictly requires a defined wavevector ($\mathbf{k} \neq 0$) for phase matching. Because $\mathbf{k}$ determines the Poynting vector $\mathbf{S}$, which identically determines T_{0i}^{EM}, the quantum mechanical condition for electromagnetic absorption is inextricably linked to the tensor condition for gravitation. The photon cannot satisfy the QED requirement for absorption without simultaneously satisfying the GR requirement for spacetime curvature.

9 The Structure of GEM in Light

9.1 Three Components, Three Geometries

A light wave propagating in direction $\hat{\mathbf{k}}$ exhibits three distinct geometric behaviors:

Component	**Geometry**	**Physical Meaning**
$\mathbf{E}$ (Electric)	Transverse oscillation $\perp$ to $\hat{\mathbf{k}}$	Force on stationary charges
$\mathbf{B}$ (Magnetic)	Transverse oscillation $\perp$ to E and $\hat{\mathbf{k}}$	Force on moving charges
$\mathbf{G}$ (Gravitational)	Longitudinal metric tension along $\hat{\mathbf{k}}$	Tensional force on mass-energy; directed momentum

Table 2: The three geometric components of a GEM wave.

The gravitational component requires careful geometric description. While $\mathbf{E}$ and $\mathbf{B}$ oscillate in transverse planes as vector fields, the energy of the wave flows longitudinally via the Poynting vector ($\mathbf{S}$). This longitudinal energy-momentum flux creates a metric tension along the propagation axis and a radial attraction toward the beam from all perpendicular directions. Note: this is not a longitudinal gravitational *wave* (gravitational waves are strictly transverse); it is a longitudinal *metric tension* created by the directed energy flux.

A photon traveling in the $+Z$ direction carries energy forward. That energy creates a gravitational well toward the beam, pulling nearby test masses toward the photon's path.

9.2 Why Gravity Appears Absent in Light

If light carries gravity, why do electromagnetic detectors fail to register it?

Geometric Mismatch: Antennas and photoreceptors are designed to couple with transverse dipole oscillations. The gravitational component creates a radial, tensor-based attraction, a completely different geometric signature that electromagnetic instruments are physically blind to.

Extreme Suppression: At the quantum scale, the ratio of electric to gravitational force exceeds 40 orders of magnitude. This suppression is not arbitrary. The metric stiffness $c^4/(16\pi G) \approx 2.41 \times 10^{42}$ N means the vacuum is extraordinarily rigid: spacetime resists curvature with enormous force. This rigidity ensures that the gravitational component, while mandatory for absorption (Section VII.C), operates at a scale far below electromagnetic dynamics. The G component need only be nonzero to satisfy the absorption condition ($T^{\mathrm{EM}}_{0i} \neq 0$) and enable phase synchronization with the target wavefunction (Section VIII.B); it does not need to be large. The vacuum's stiffness keeps it at the minimum required magnitude: present enough to carry directed momentum and phase, negligible compared to the electromagnetic forces that govern atomic structure.

Null Causal Isolation: As established in Section V, parallel photons on null geodesics are causally blind to one another's metric tension. Only in specific configurations (light meeting mass, or anti-parallel beams) does the gravitational component manifest without causal isolation.

Gravity in light is present. It is geometrically orthogonal, heavily suppressed by the vacuum's metric stiffness, and functionally constrained to its essential role: enabling the directed, phase-coherent momentum transfer that makes photon-matter interaction possible.

10 Matter as Frozen GEM: Compactification and the Dirac Factorization

The deepest evidence for the GEM framework lies in the absolute interconvertibility of matter and radiation, observed macroscopically in pair production near a heavy nucleus:

$$\gamma + Z \to e^+ + e^- + Z\,. \tag{11}$$

A photon above 1.022 MeV, when interacting with a background nuclear field to conserve momentum, converts directly into an electron and a positron. If matter and light were fundamentally different substances, this would be an impossible transmutation. They are, instead, different topological configurations of the exact same underlying GEM structure.

10.1 Compactification and Unitary Phase Rotation

The conversion of radiation to matter represents a transition from **spatial propagation** to **momentum confinement**. For a freely propagating photon, the momentum wavevector is real and longitudinal ($p_z = \hbar k$). To form a bound particle, the wavefunction must satisfy closed boundary conditions. The wavevector undergoes a unitary rotation in the complex plane:

$$k \to i\kappa\,. \tag{12}$$

This 90-degree phase rotation converts propagating solutions (e^{ikz}) into confined, localized solutions ($e^{-\kappa r}$). The longitudinal propagation energy of the GEM wave is geometrically locked, manifesting macroscopically as rest mass ($m = E/c^2$). What was forward motion becomes internal circulation.

10.2 The Boson-to-Fermion Transition: The Dirac Factorization

A critical objection arises: photons are commuting spin-1 bosons; electrons are anti-commuting spin-1/2 fermions. How can a unitary rotation convert a vector field into a Grassmann variable without violating quantum mechanics?

The boson-to-fermion transition is not an arbitrary topological assumption; it is an algebraic necessity of confinement.

Free electromagnetic radiation satisfies the massless second-order wave equation ($\partial_\mu \partial^\mu A^\mu = 0$). When the wavevector rotates ($k \to i\kappa$) during pair production, the field does not merely acquire mass (which would yield the bosonic Proca equation); rather, the topological compactification decomposes the spin-1 vector geometry into its fundamental spin-1/2 irreducible representations (the specific geometric manifold governing this dimensional projection is derived in [12]). To maintain a positive-definite probability density for these confined states, the second-order wave mechanics must obey the massive Klein–Gordon constraint:

$$(\partial_\mu \partial^\mu + m^2)\phi = 0\,. \tag{13}$$

To achieve first-order time evolution (the core requirement of unitary quantum mechanics), this second-order differential operator must be factorized. As P.A.M. Dirac proved in 1928 [9], factorizing ($\partial_\mu \partial^\mu + m^2$) is impossible with standard commuting numbers. It strictly requires the introduction of mutually anticommuting algebraic objects: the Dirac gamma matrices (γ^μ).

The factorization of the confined GEM wave directly yields the Dirac equation:

$$(i\gamma^\mu \partial_\mu - m)\psi = 0\,. \tag{14}$$

Because the gamma matrices operate on 4-component bispinors, the resulting confined wavefunction ψ is mathematically forced to exhibit spin-1/2 statistics. The 720° phase geometry of fermions is not a new property injected into the field; it is the unavoidable geometric consequence of linearizing a massive, confined vector field. The single spin-1 photon becomes two spin-1/2 fermions ($\gamma \to e^+ + e^-$), perfectly conserving total angular momentum within the pair's combined spin and orbital state, while satisfying the topological dictates of the Dirac algebra.

10.3 Inertia as Topological Confinement

Mass-energy equivalence ($E = mc^2$) acquires rigid geometric meaning through this framework. Rest mass is not a separate substance added to the field. When a photon converts to an electron-positron pair:

- Spatial propagation $(x, y, z) \to$ converts to momentum confinement (p_x, p_y, p_z)
- Longitudinal propagation $p_z \to$ rotates into confined spin angular momentum
- Neutral vector fields ($\partial_\mu \partial^\mu A^\mu = 0$) $\to$ factorize into charged Dirac spinors ($(i\gamma^\mu \partial_\mu - m)\psi = 0$)
- The Poynting vector (G component) $\to$ becomes the particle's gravitational mass

- GEM radiation → becomes GEM matter

Particles are not made of something other than light. Matter is frozen GEM: radiation whose propagation vector has undergone a topological compactification, locking its energy and momentum into a recursive, localized Dirac state.

11 Falsifiable Predictions and Modularity

A framework that cannot be falsified is not physics. However, much like the Standard Model, the overarching unified geometry underlying this thesis (the OMEGA 137 architecture [12]) is highly modular. In such architectures, an experimental discrepancy typically indicates the presence of a scale-dependent symmetry break or the need for a localized perturbative connection, rather than a failure of the entire mathematical foundation. Within this modular structure, the specific GEM triad presented here makes strict, testable predictions:

Prediction	**Test**	**Current Status**
$c_{\text{gravity}} = c_{\text{light}}$ exactly	LIGO + EM counterpart timing	Consistent: $\Delta c/c < 10^{-15}$ [7, 8]
Parallel light beams show zero mutual gravitational attraction	Precision beam experiments	Consistent (TEP, 1931)
Anti-parallel light beams show net gravitational attraction	Precision beam experiments	Predicted; interaction $\propto T_{00} + T_{zz}$, direct measurement pending

Table 3: Falsifiable predictions of the GEM framework.

Any observation of perfectly parallel light beams gravitationally attracting would falsify the null causal isolation of the GEM tensor geometry. Finally, any confirmed vacuum divergence between c_{gravity} and c_{light} at extreme precision limits would not necessarily invalidate the single continuous medium, but it would require a scale-dependent perturbative correction to the macroscopic phase-lock between the electromagnetic and gravitational parameters.

12 Conclusions

By identifying the electromagnetic stress-energy tensor as the rank-2 bridge to spacetime curvature, this framework unifies ε_0, μ_0, and G as the continuous properties of a single vacuum medium. The electric field exerts force on stationary charges. The magnetic field exerts force on moving charges. The gravitational component exerts tensional force on mass-energy. These three constitute the GEM triad.

The gravitational component of light is not a passive byproduct of energy; it is required by the absorption math itself. The quantum absorption Hamiltonian requires $\mathbf{k} \neq 0$; this is identically the condition $T^{\text{EM}}_{0i} \neq 0$ that sources spacetime curvature. The same mathematical condition that permits a photon to be absorbed by an atom guarantees that it gravitates.

The conversion of matter to light (and light to matter) is a topological compactification. Pair production compactifies a freely propagating GEM wave from spatial propagation (x, y, z) into confined momentum phase space (p_x, p_y, p_z), with the Dirac factorization providing the algebraic necessity for the boson-to-fermion transition. Matter is frozen GEM.

Light does not merely carry energy that happens to gravitate.

Light is GEM.

References

[1] Maxwell, J.C. (1865). "A Dynamical Theory of the Electromagnetic Field." *Phil. Trans. R. Soc. Lond.* **155**, 459–512.

[2] Einstein, A. (1905). "Zur Elektrodynamik bewegter Körper." *Ann. Phys.* **17**, 891–921.

[3] Einstein, A. (1915). "Die Feldgleichungen der Gravitation." *Sitz. Preuss. Akad. Wiss.* 844–847.

[4] Dyson, F.W., Eddington, A.S., Davidson, C. (1920). "A Determination of the Deflection of Light by the Sun's Gravitational Field." *Phil. Trans. R. Soc. A* **220**, 291–333.

[5] Pound, R.V., Rebka, G.A. (1960). "Apparent Weight of Photons." *Phys. Rev. Lett.* **4**, 337–341.

[6] Tolman, R.C., Ehrenfest, P., Podolsky, B. (1931). "On the Gravitational Field Produced by Light." *Phys. Rev.* **37**, 602–615.

[7] Abbott, B.P. et al. (2017). "GW170817: Observation of Gravitational Waves from a Binary Neutron Star Inspiral." *Phys. Rev. Lett.* **119**, 161101.

[8] Abbott, B.P. et al. (2017). "Gravitational Waves and Gamma-Rays from a Binary Neutron Star Merger." *Astrophys. J. Lett.* **848**, L13.

[9] Dirac, P.A.M. (1928). "The Quantum Theory of the Electron." *Proc. R. Soc. Lond. A* **117**(778), 610–624.

[10] Poynting, J.H. (1884). "On the Transfer of Energy in the Electromagnetic Field." *Phil. Trans. R. Soc. Lond.* **175**, 343–361.

[11] Peskin, M.E., Schroeder, D.V. (1995). *An Introduction to Quantum Field Theory.* Westview Press.

[12] Lehew, J. (2026). "Zero Degrees of Freedom: The Geometric Unification of Gravity, Time Dilation, and the Fine-Structure Constant (α)." OMEGA 137 Framework. ORCID: 0009-0007-5895-7787.